NovaTech Systems — Engineering Handbook

NovaTech Systems • Internal Knowledge Base Document

Field	Value
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Status	Active

Purpose and scope

Defines the standard software-engineering workflow from branch creation through review, CI, testing, deployment, and incident response.

All names, systems, processes, dates, limits, and examples in this document are fictional and exist only for the NovaTech Systems RAG knowledge-base exercise.

1. Engineering Principles

- Prefer simple, observable, maintainable solutions over unnecessary complexity.
- Design for failure in distributed systems and make operational behavior visible through logs and metrics.
- Treat security and privacy as design requirements, not final-stage checks.
- Automate repetitive validation through CI/CD.
- Document decisions that future engineers would otherwise have to rediscover.

2. Git Workflow

Branch naming

Purpose	Pattern	Example
New functionality	feature/-	feature/workflow-sla-rules
Defect correction	bugfix/-	bugfix/token-refresh-timeout
Urgent production correction	hotfix/-	hotfix/novaflow-queue-failure
Release branch	release/-	release/2026.10

Changes should normally be made on a short-lived feature, bugfix, or hotfix branch and merged through a pull request. Direct pushes to protected branches are not part of the standard workflow.

3. Pull Requests

Minimum requirements for a pull request are: at least one reviewer, passing automated tests, passing static analysis, and passing required security checks.

- Describe the problem, approach, and meaningful behavior changes.
- Link the relevant issue or work item.
- Include tests for changed behavior and regression-prone logic.
- Call out database, API, configuration, migration, or rollout considerations.
- Do not approve a PR when mandatory CI checks are failing unless the exception is formally documented and authorized.

4. Code Reviews

Reviewers should evaluate correctness, security, performance, maintainability, test coverage, API compatibility, error handling, logging, naming, and documentation. See NT-ENG-002 for detailed review guidance.

5. Testing

- Unit tests validate isolated business logic.
- Integration tests validate interactions with persistence, messaging, external services, or other meaningful boundaries.
- API or contract tests should be used where compatibility is important.
- Production-impacting changes should include a rollback or mitigation consideration.

6. CI/CD

The standard delivery path is pull request → CI → automated tests → security scan → build → staging → QA validation → production deployment. Deployment details are maintained with the deployment process owned by engineering/SRE.

7. Deployment

Production deployments normally occur Tuesday and Thursday from 10:00 AM to 3:00 PM IST. Emergency production deployments can occur outside this window when incident approval is obtained.

8. Logging and Monitoring

- Logs should be structured where practical and must not expose passwords, API keys, access tokens, or unnecessary personal data.
- Services should expose useful health and operational signals appropriate to their role.
- Alerts should be actionable and tied to a clear response or escalation path.

9. Incident Management

Engineering teams participate in incidents according to NT-SRE-001. During a production incident, stabilize customer impact first, preserve evidence, communicate clearly, and document follow-up actions.

10. Version History

Version	Effective date	Status	Change summary
1.8	2025-10-01	Historical	Earlier engineering workflow and review baseline.
2.0	2026-06-15	Active	Clarified mandatory CI/security gates, branch conventions, and deployment expectations.

11. Related Documents

- NT-ENG-002 — Code Review Guidelines
- NT-SEC-001 — Security Policy
- NT-IT-001 — IT Support Guide
- NT-SRE-001 — Incident Management Runbook
- NT-HR-005 — Employee Onboarding Guide